

Ryan Cho

Senior Software Engineer | Backend Systems & Integrations

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PROFESSIONAL SUMMARY

Software engineer with 7+ years building and operating production backend systems across finance and SaaS. Deep experience in TypeScript/Node.js services, third-party API integrations, and queue-based workflows with PostgreSQL on AWS. Proven track record delivering idempotent, observable pipelines, LLM-assisted tooling, and secure data exchange at scale. Strong ownership from design through deployment, monitoring, and incident resolution.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and operated **TypeScript/Node.js integration services** moving portfolio and operational data between internal platforms and external ERP systems, supporting bidirectional sync across 15+ institutional client accounts.
- Built **queue-based workers** with PostgreSQL-backed job stores featuring idempotency keys, exponential backoff retries, dead-letter queues, and correlation IDs for full audit replay.
- Designed **typed REST and OAuth2 API contracts** against third-party vendor endpoints, adding rate limiting and webhook ingestion to keep sync jobs replay-safe under burst traffic.
- Deployed services on **AWS with Docker and Kubernetes** via GitHub Actions CI/CD, owning releases, rollbacks, and post-deploy validation within defined operational SLAs.
- Instrumented pipelines with **structured logging, metrics, and alerting** to surface data accuracy drift early in multi-tenant ingestion workflows before customer impact.
- Contributed to **LLM-assisted onboarding tooling** that reduced manual mapping work for new customer integrations by generating validation scripts from sample ERP exports.
- Optimized **PostgreSQL schemas and query paths** for high-volume ingestion tables, improving throughput for the largest tenant nightly reconciliation workloads.
- Partnered with product and operations stakeholders to **troubleshoot production incidents**, document root causes, and harden services against edge cases in upstream API behavior.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **production backend services** in TypeScript and Python supporting trade operations, regulatory reporting, and internal data pipelines across distributed banking platforms.
- Integrated internal systems with **third-party REST and SOAP APIs**, implementing OAuth2 flows and strict typed schemas to exchange market and counterparty data securely.
- Implemented **message-queue processing** with retry policies, dead-letter handling, and backpressure controls to keep batch reconciliation resilient during upstream outages.
- Operated **PostgreSQL and relational data stores** at scale, tuning indexes and batch writes for reporting workloads spanning millions of daily transaction records.
- Established **observability standards** including centralized logging, trace correlation, and dashboard alerts used by on-call teams across 3 regional production environments.
- Applied **secure development practices** including credential rotation, least-privilege IAM, and input validation to integration layers handling sensitive financial data.
- Collaborated across technology, risk, and business units to **deliver integration milestones** on quarterly release cycles while maintaining audit-ready change documentation.
- Mentored junior analysts on **API design, testing strategies, and incident response**, improving review turnaround and reducing rework on integration deliverables.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **Node.js HTTP APIs** supporting product inventory and order workflows, integrating partner services through REST endpoints and webhook callbacks.
- Implemented **background job processors** with retry logic and idempotent handlers to synchronize catalog updates without duplicate side effects across 5 downstream services.
- Developed **PostgreSQL data models** and migration scripts for feature releases, ensuring backward-compatible schema changes across staging and production.
- Wrote integration tests and **CI pipelines in GitHub Actions** to catch contract regressions before deployment to cloud-hosted environments.
- Supported production troubleshooting by adding **request logging and error tracking**, shortening diagnosis time for customer-reported sync failures.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Delivered **responsive web applications** and admin tooling for nonprofit programs, connecting front-end forms to backend services over REST APIs.
- Implemented **server-side data validation** and authentication checks to protect user submissions and align with secure development practices.
- Configured **deployment pipelines** and environment settings for remote delivery, enabling stable releases during a compressed 10-week project timeline.
- Collaborated with stakeholders to **translate requirements into technical tasks**, shipping iterative feature updates ahead of program launch deadlines.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML/CSS, Responsive UI

Backend: Node.js, TypeScript, Python, Elixir, REST APIs, SOAP, OAuth2, Webhooks, Queue Workers, Idempotent Job Design

Databases: PostgreSQL, SQL, Schema Design, Query Optimization, Migrations, Relational Data Modeling

Cloud & DevOps: AWS, Docker, Kubernetes, GitHub Actions, CI/CD, Infrastructure as Code, Secure Deployment

Monitoring & Tools: Structured Logging, Metrics & Alerting, Distributed Tracing, Observability, Incident Triage, Datadog, Grafana

Workflow & PM: Agile/Scrum, Technical Documentation, Code Review, Cross-Functional Collaboration, Stakeholder Communication